

BCA 4th Semester – Scripting Language

Assignment Answers (Complete Q1 to Q14)

1. What do you understand by JavaScript?

- JavaScript is a client-side scripting language used to create dynamic, interactive, and responsive web pages. It can modify HTML, change CSS, handle events, and communicate with servers.

2. Program to print smallest and largest numbers from 15 elements of an array.

- Example:

```
let arr = [12, 5, 18, 25, 3, 9, 15, 30, 2, 19, 7, 11, 6, 28, 1];
let smallest = arr[0];
let largest = arr[0];
for (let i = 1; i < arr.length; i++) {
    if (arr[i] < smallest) smallest = arr[i];
    if (arr[i] > largest) largest = arr[i];
}
console.log('Smallest:', smallest);
console.log('Largest:', largest);
```

3. Explain the concept and importance of the DOM.

- The Document Object Model (DOM) is a tree-like structure that represents an HTML document. It allows JavaScript to access, modify, add, or remove elements dynamically, making pages interactive.

4. Explain the structure of the DOM.

- The DOM is structured like a tree:

Document → HTML → Head → Body → Elements.

It consists of nodes such as document nodes, element nodes, attribute nodes, and text nodes.

5. Why do we need dialog boxes? Explain types with examples.

- Dialog boxes display messages or receive input.

Types:

- **alert()** – Displays a simple message.

Example: `alert('Hello!');`

- **confirm()** – Asks the user to confirm an action, returning true if OK is clicked and false if Cancel.

Example: `confirm('Are you sure?');`

- **prompt()** – Prompts the user to enter some input. Returns the input as a string.

Example: `prompt('Enter name:');`

6. Explain HTTP function. Explain GET & POST.

- HTTP is a communication protocol used for client-server communication.

- GET is an HTTP method used to request or retrieve data from the server.

It sends data through the URL and is mainly used for reading or viewing information.

- POST is an HTTP method used to send data to the server for processing or storing.

It sends data through the request body and is mainly used for submitting forms and uploading data.

7. Difference between GET and POST Methods (Table Included)

Aspect	GET	POST
Purpose	The GET method is used to request or retrieve data from the server.	The POST method is used to send or submit data to the server.
Data Location	GET sends data in the URL as query parameters.	POST sends data in the request body.
Security	GET is less secure because data is visible in the URL.	POST is more secure because data is not shown in the URL.

Data Size	GET supports limited data length because URL length is restricted.	POST supports unlimited data because it sends data in the body.
Use Case	GET is used for reading or requesting information without side effects.	POST is used for submitting forms, uploading files, and updating data.

8. What is the use of jQuery? Write program to hide/show a div.

- jQuery simplifies JavaScript tasks like DOM manipulation and event handling.

Example:

```
$('#hide').click(function(){ $('#box').hide(); });
$('#show').click(function(){ $('#box').show(); });
```

9. Explain Ajax.

- Ajax allows webpages to request data from the server without reloading, making them faster and more dynamic.

10. Why is Ajax required in web development?

- Ajax improves speed, updates only parts of a page, enhances user experience, and supports background data loading.

11. What do you understand about CMS? Steps to create submenu in Joomla.

- A CMS manages website content without coding.

Steps in Joomla:

1. Login admin.
2. Go to Menus → Main Menu.
3. Add new menu item.
4. Choose parent item.
5. Save to create submenu.

12. Discuss different types of events and how to handle them in JavaScript.

- Types: Mouse events, keyboard events, form events, window events.

Example:

```
document.getElementById('btn').onclick = function(){ alert('Clicked!'); };
```

13. Explain form validation using JavaScript.

- Form validation checks user inputs before submission.

Example:

```
if(document.getElementById('name').value==""){ alert('Enter name'); return false; }
```

14. Explain structure and features of JavaScript.

Structure:

- Variables: Used to store values.
- Operators: Perform calculations and comparisons.
- Conditionals: Execute code based on conditions.
- Loops: Repeat tasks until a condition is met.
- Functions: Reusable code blocks.
- Objects: Store data as key-value pairs.
- Events: React to actions like clicks.
- DOM Manipulation: Modify webpage content dynamically.

Features:

- Lightweight: Runs efficiently in browsers.
- Object-Based: Uses objects for structure.
- Platform-Independent: Works on any browser.
- Event Handling: Responds to user actions.
- Asynchronous: Supports non-blocking operations.
- HTML/CSS Integration: Works with web technologies.
- Dynamic Typing: Variables can change types.
- Fast Execution: Runs quickly using browser engine.